ROBOTICS 4/M

TOPIC 9: DYNAMICS EQUATIONS FOR MULTIPLE DEGREE OF FREEDOM ROBOTS

9.1 LAGRANGIAN APPROACH

As the number of degrees of freedom increases the complexity of the Equations of Motion increases too. Consequently, the derivation of these equations becomes more complicated and long. However, using Lagrangian Mechanics they can be derived by calculating the kinetic and potential energies of the links and joints, and then developing the Lagrangian derivatives. Firstly, we will consider the Kinetic and Potential Energies derivations separately [1,2].

9.2 KINETIC ENERGY

Kinetic energy is the energy used to move an object. For a rigid body the kinetic energy has both translational and rotational components, as shown here [1]:

$$K = \tfrac{1}{2} m V^2 + \tfrac{1}{2} I \omega^2 \tag{9.1}$$

Where m is the mass components of the system and I represents the inertia of the system. In order to calculate this equation, a suitable expression for the velocity of the robot's links can be derived.

VELOCITY CALCULATION

The velocity of the link is obtained by differentiating the position of the related joint frames relative to a fixed reference frame [1,2]. Using r_i as the position of a point on the link i, then the position of this point relative to the universally fixed reference frame, p_i, is obtained by using the transformation matrix from the fixed frame to the joint associated with the link being considered:

$$p_i = {}^U T_i r_i \tag{9.2}$$

Therefore differentiating Equation (9.2) with respect to time yields the velocity relationship of the point i.e.

$$V_i = \frac{dp_i}{dt} = \frac{d}{dt}\left({}^U T_i r_i\right) \tag{9.3}$$

The velocity of the point is related to the combined velocities of the joints i.e. $\dot{q}_1, \dot{q}_2, \cdots \dot{q}_6$ for a six degree of freedom robot. Therefore the velocity relationship in Equation (9.3) is also related to the cumulative effect of the velocities of the joints. This can be introduced into the relationship by the partial derivative of the transformation matrix with respect to generic joint variables, q_i [1] i.e.

$$V_i = \frac{d}{dt}\left({}^U T_i r_i\right) = \sum_{j=1}^{i}\left(\frac{\partial {}^U T_i}{\partial q_i}\frac{dq_j}{dt}\right) r_i \tag{9.4}$$

Now we must consider the derivation of the transformation matrix derivative. Previously, we saw that the position and orientation is defined by the kinematic equations for the robot configuration being considered. For this derivation the Denavit-Hartenberg transformation matrices, A_i, will be used to derive the velocity relationships for points along the robot's links [1,2]. The transformation for a 6 degree of freedom robot is

$${}^U T_H = A_1 A_2 A_3 A_4 A_5 A_6 \tag{9.5}$$

The derivative of this transformation depends on the type of joint being considered.

PRISMATIC JOINT

The joint variable for a prismatic joint is the linear displacement d_i and thus the derivative of the transformation matrix relative to that joint must be with respect to this joint variable [1]:

$$\frac{\partial A_i}{\partial d_i} = \frac{\partial}{\partial d_i}\begin{bmatrix} C\theta_i & -S\theta_i C\alpha_i & S\theta_i S\alpha_i & a_i C\theta_i \\ S\theta_i & C\theta_i C\alpha_i & -C\theta_i S\alpha_i & a_i S\theta_i \\ 0 & S\alpha_i & C\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{9.6}$$

Due to the similarity between the derivative and the original transformation matrices, the matrix in (9.6) can be split into a constant matrix Q_i and the transformation matrix A_i. This gives [1]:

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} C\theta_i & -S\theta_i C\alpha_i & S\theta_i S\alpha_i & a_i C\theta_i \\ S\theta_i & C\theta_i C\alpha_i & -C\theta_i S\alpha_i & a_i S\theta_i \\ 0 & S\alpha_i & C\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{9.7}$$

or

$$\frac{\partial A_i}{\partial d_i} = Q_i A_i \tag{9.8}$$

REVOLUTE JOINT

The joint variable for a revolute joint is the angular displacement θ_i and thus the derivative of the transformation matrix relative to that joint must be with respect to this joint variable [1]:

$$\frac{\partial A_i}{\partial \theta_i} = \frac{\partial}{\partial \theta_i} \begin{bmatrix} C\theta_i & -S\theta_i C\alpha_i & S\theta_i S\alpha_i & a_i C\theta_i \\ S\theta_i & C\theta_i C\alpha_i & -C\theta_i S\alpha_i & a_i S\theta_i \\ 0 & S\alpha_i & C\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -S\theta_i & -C\theta_i C\alpha_i & C\theta_i S\alpha_i & a_i S\theta_i \\ C\theta_i & -S\theta_i C\alpha_i & S\theta_i S\alpha_i & a_i C\theta_i \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{9.9}$$

Due to the similarity between the derivative and the original transformation matrices, the matrix in (9.9) can be split into a constant matrix Q_i and the transformation matrix A_i. This gives [1]:

$$\begin{bmatrix} -S\theta_i & -C\theta_i C\alpha_i & C\theta_i S\alpha_i & a_i S\theta_i \\ C\theta_i & -S\theta_i C\alpha_i & S\theta_i S\alpha_i & a_i C\theta_i \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} C\theta_i & -S\theta_i C\alpha_i & S\theta_i S\alpha_i & a_i C\theta_i \\ S\theta_i & C\theta_i C\alpha_i & -C\theta_i S\alpha_i & a_i S\theta_i \\ 0 & S\alpha_i & C\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{9.10}$$

or

$$\frac{\partial A_i}{\partial \theta_i} = Q_i A_i \tag{9.11}$$

Therefore the Q_i matrices are as follows:

$$Q_i(prismatic) = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \; Q_i(revolute) = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{9.12}$$

TRANSFORMATION MATRIX DERIVATIVE AND VELOCITY

The previous transformation matrices derivatives in (9.6) and (9.10) have been derived relative to the frame of reference assigned to the joint being considered [1]. In Equation (9.4) the derivative is of the transformation relative to the fixed reference frame. This is obtained by considered all the transformations from the fixed to the joint frame:

$$U_{ij} = \frac{\partial {}^U T_i}{\partial q_j} = \frac{\partial \left(A_1 A_2 \cdots A_j \cdots A_i \right)}{\partial q_j} = A_1 A_2 \cdots Q_j A_j \cdots A_i \quad j \le i \tag{9.13}$$

This shows that only the transformation of the joint being considered is affected by the differentiation. Higher order derivatives can be calculated as follows:

$$U_{ijk} = \frac{\partial U_{ij}}{\partial q_k} \tag{9.14}$$

Equation (9.13) can be substituted into the velocity equation (9.4) thus:

$$V_i = \frac{d}{dt}\left({}^U T_i r_i \right) = \sum_{j=1}^{i} \left(\frac{\partial {}^U T_i}{\partial q_i} \frac{dq_j}{dt} \right) r_i = \sum_{j=1}^{i} \left(U_{ij} \frac{dq_j}{dt} \right) r_i \tag{9.15}$$

KINETIC ENERGY CALCULATION

Since the velocity V_i represents the velocity in 3 dimensions it is vector with components $\dot{x}_i, \dot{y}_i, \dot{z}_i$. It follows that the kinetic energy of this point on the link i is [1]:

$$dK_i = \tfrac{1}{2} V_i V_i^T dm_i = \tfrac{1}{2}\left(\dot{x}_i^{\,2} + \dot{y}_i^{\,2} + \dot{z}_i^{\,2}\right) dm_i \tag{9.16}$$

Since V_i is a vector the following relationship holds:

$$V_i V_i^T = \begin{bmatrix} \dot{x}_i \\ \dot{y}_i \\ \dot{z}_i \end{bmatrix} \begin{bmatrix} \dot{x}_i & \dot{y}_i & \dot{z}_i \end{bmatrix} = \begin{bmatrix} \dot{x}_i^{\,2} & \dot{x}_i\dot{y}_i & \dot{x}_i\dot{z}_i \\ \dot{y}_i\dot{x}_i & \dot{y}_i^{\,2} & \dot{y}_i\dot{z}_i \\ \dot{z}_i\dot{x}_i & \dot{z}_i\dot{y}_i & \dot{z}_i^{\,2} \end{bmatrix} \tag{9.17}$$

and [1]

$$Trace\left(V_i V_i^T\right) = Trace\begin{bmatrix} \dot{x}_i^{\,2} & \dot{x}_i\dot{y}_i & \dot{x}_i\dot{z}_i \\ \dot{y}_i\dot{x}_i & \dot{y}_i^{\,2} & \dot{y}_i\dot{z}_i \\ \dot{z}_i\dot{x}_i & \dot{z}_i\dot{y}_i & \dot{z}_i^{\,2} \end{bmatrix} = \dot{x}_i^{\,2} + \dot{y}_i^{\,2} + \dot{z}_i^{\,2} \tag{9.18}$$

Substituting this into Equation (9.16) yields:

$$dK = \tfrac{1}{2}\left(\dot{x}_i^{\,2} + \dot{y}_i^{\,2} + \dot{z}_i^{\,2}\right) dm = \tfrac{1}{2} Trace\left(V_i V_i^T\right) dm \tag{9.19}$$

This can them be extended using the velocity equation provided in Equation (9.15):

$$dK_i = \tfrac{1}{2} Trace\left(V_i V_i^T\right) dm_i = \tfrac{1}{2} Trace\left[\left(\sum_{p=1}^{i}\left(U_{ip}\frac{dq_p}{dt}\right) r_i\right)\cdot\left(\sum_{r=1}^{i}\left(U_{ir}\frac{dq_r}{dt}\right) r_i\right)^T\right] dm_i \tag{9.19}$$

Here p and r are the different joint numbers. Integrating the above equation over the length of the link provides the kinetic energy expression for the link i.e.

$$K_i = \int dK_i = \tfrac{1}{2} Trace\left[\sum_{p=1}^{i}\sum_{r=1}^{i} U_{ip}\left(\int r_i r_i^T dm_i\right) U_{ir}^{\,T} \frac{dq_p}{dt}\frac{dq_r}{dt}\right] \tag{9.20}$$

The integral term within this equation is called the pseudo inertia matrix which can be written as [1]:

$$I_i = \begin{bmatrix} \left(-I_{xx} + I_{yy} + I_{zz}\right)/2 & I_{xy} & I_{xz} & m_i x_i \\ I_{xy} & \left(I_{xx} - I_{yy} + I_{zz}\right)/2 & I_{yz} & m_i y_i \\ I_{xz} & I_{yz} & \left(I_{xx} + I_{yy} - I_{zz}\right)/2 & m_i z_i \\ m_i x_i & m_i y_i & m_i z_i & m_i \end{bmatrix} \tag{9.21}$$

This gives the kinetic energy equation for the link as:

$$K_i = \tfrac{1}{2}\sum_{p=1}^{i}\sum_{r=1}^{i} Trace\left(U_{ip} I_i U_{ir}^{\,T}\right)\frac{dq_p}{dt}\frac{dq_r}{dt} \tag{9.22}$$

The total kinetic energy for the links of the robot manipulator is

$$K_{links} = \tfrac{1}{2}\sum_{i=1}^{n}\sum_{p=1}^{i}\sum_{r=1}^{i} Trace\left(U_{ip} I_i U_{ir}^{\,T}\right)\frac{dq_p}{dt}\frac{dq_r}{dt} \tag{9.23}$$

The kinetic energy of the actuators has to be included. This has the following form:

$$K_{actuators} = \tfrac{1}{2}\sum_{i=1}^{n} I_{i(actuators)}\left(\frac{dq_i}{dt}\right)^2 \tag{9.24}$$

The total kinetic energy for the manipulator is

$$K = K_{links} + K_{actuators} = \tfrac{1}{2}\sum_{i=1}^{n}\sum_{p=1}^{i}\sum_{r=1}^{i} Trace\left(U_{ip} I_i U_{ir}{}^T\right)\frac{dq_p}{dt}\frac{dq_r}{dt} + \tfrac{1}{2}\sum_{i=1}^{n} I_{i(actuators)}\left(\frac{dq_i}{dt}\right)^2 \tag{9.25}$$

9.3 Potential Energy

The potential energy is the energy stored within an object. For a rigid body the potential energy has the form shown here:

$$P = mgh \tag{9.26}$$

Where m is the mass components of the system, g is the gravity component and h is the height of the system. In order to calculate the potential energy in each link the position equation (9.2) is used to represent h. This gives the following equation for the potential energy in link i [1,2]:

$$P = \sum_{i=1}^{n} P_i = \sum_{i=1}^{n}\left(-m_i \mathbf{g}^T \left({}^U T_i r_i\right)\right) \tag{9.27}$$

where $\mathbf{g}^T = [g_x\ g_y\ g_z\ 0]$ is the gravity components of the link in three dimensions and r_i is the location of the centre of mass of the link relative to the joint reference frame associated with the link.

9.4 The Lagrangian

The Lagrangian for a multiple degree of freedom manipulator is formed by combining Equations (9.25) and (9.27):

$$L = K - P = \tfrac{1}{2}\sum_{i=1}^{n}\sum_{p=1}^{i}\sum_{r=1}^{i} Trace\left(U_{ip} I_i U_{ir}{}^T\right)\frac{dq_p}{dt}\frac{dq_r}{dt} + \tfrac{1}{2}\sum_{i=1}^{n} I_{i(actuators)}\left(\frac{dq_i}{dt}\right)^2 - \sum_{i=1}^{n}\left(-m_i \mathbf{g}^T \left({}^U T_i r_i\right)\right) \tag{9.28}$$

This equation is used to define the equations of motion for the robot [1].

9.5 The Equations of Motion for a Robot Manipulator

The process of differentiating the Lagrangian yields the following equations of motion for a particular joint/link i [1]:

$$T_i = \sum_{j=1}^{n} D_{ij}\frac{d^2 q_j}{dt^2} + I_{i(actuators)}\frac{d^2 q_i}{dt^2} + \sum_{j=1}^{n}\sum_{k=1}^{n} D_{ijk}\frac{dq_j}{dt}\frac{dq_k}{dt} + D_i \tag{9.29}$$

where

$$D_{ij} = \sum_{p=\max(i,j)}^{n} Trace\left(U_{pj} I_p U_{pi}{}^T\right) \tag{9.30}$$

and

$$D_{ijk} = \sum_{p=\max(i,j)}^{n} Trace\left(U_{pjk} I_p U_{pik}{}^T\right) \tag{9.31}$$

and

$$D_i = \sum_{p=i}^{n}\left(-m_p \mathbf{g}^T U_{pi} r_p\right) \tag{9.32}$$

In Equation (9.29) the first part is the inertia term, the second is the actuator inertia term, the third part is the Coriolis terms, and the last term is the gravity effect [1]. This equation expands for a robot manipulator with only revolute joints as follows:

$$
\begin{aligned}
T_i = {} & D_{i1}\ddot{\theta}_1 + D_{i2}\ddot{\theta}_2 + D_{i3}\ddot{\theta}_3 + D_{i4}\ddot{\theta}_4 + D_{i5}\ddot{\theta}_5 + D_{i6}\ddot{\theta}_6 + I_{i(actuators)}\ddot{\theta}_1 \\
& + D_{i11}\dot{\theta}_1^{\,2} + D_{i22}\dot{\theta}_2^{\,2} + D_{i33}\dot{\theta}_3^{\,2} + D_{i44}\dot{\theta}_4^{\,2} + D_{i55}\dot{\theta}_5^{\,2} + D_{i66}\dot{\theta}_6^{\,2} \\
& + D_{i12}\dot{\theta}_1\dot{\theta}_2 + D_{i13}\dot{\theta}_1\dot{\theta}_3 + D_{i14}\dot{\theta}_1\dot{\theta}_4 + D_{i15}\dot{\theta}_1\dot{\theta}_5 + D_{i16}\dot{\theta}_1\dot{\theta}_6 \\
& + D_{i21}\dot{\theta}_2\dot{\theta}_1 + D_{i23}\dot{\theta}_2\dot{\theta}_3 + D_{i24}\dot{\theta}_2\dot{\theta}_4 + D_{i25}\dot{\theta}_2\dot{\theta}_5 + D_{i26}\dot{\theta}_2\dot{\theta}_6 \\
& + D_{i31}\dot{\theta}_3\dot{\theta}_1 + D_{i32}\dot{\theta}_3\dot{\theta}_2 + D_{i34}\dot{\theta}_3\dot{\theta}_4 + D_{i35}\dot{\theta}_3\dot{\theta}_5 + D_{i36}\dot{\theta}_3\dot{\theta}_6 \\
& + D_{i41}\dot{\theta}_4\dot{\theta}_1 + D_{i42}\dot{\theta}_4\dot{\theta}_2 + D_{i43}\dot{\theta}_4\dot{\theta}_3 + D_{i45}\dot{\theta}_4\dot{\theta}_5 + D_{i46}\dot{\theta}_4\dot{\theta}_6 \\
& + D_{i51}\dot{\theta}_5\dot{\theta}_1 + D_{i52}\dot{\theta}_5\dot{\theta}_2 + D_{i53}\dot{\theta}_5\dot{\theta}_3 + D_{i54}\dot{\theta}_5\dot{\theta}_4 + D_{i56}\dot{\theta}_5\dot{\theta}_6 \\
& + D_{i61}\dot{\theta}_6\dot{\theta}_1 + D_{i62}\dot{\theta}_6\dot{\theta}_2 + D_{i63}\dot{\theta}_6\dot{\theta}_3 + D_{i64}\dot{\theta}_6\dot{\theta}_4 + D_{i65}\dot{\theta}_6\dot{\theta}_5 + D_i
\end{aligned}
\tag{9.33}
$$

This equation can be applied to each of the joints/links in the robot configuration and provides the equations of motion for each. Obviously, T_i represents forces for prismatic joints and torques for revolute joints.

9.6 Static Force Analysis of Robots

Static force analysis is needed for robot control, which will be dealt with later in this course. In order to create the desired force at the end effector, the controller must be able to make the joints produce the required forces and moments to accomplish the desired effect. To do this the joint forces and torques should be related to the forces and torques of the end effector. Firstly we define the forces and moments of the end effector frame [1] i.e.

$$\left[{}^{H}F\right] = \left[f_x \;\; f_y \;\; f_z \;\; m_x \;\; m_y \;\; m_z\right]^T \tag{9.34}$$

Here f_x, f_y, f_z are the forces acting along the x, y, z axes and m_x, m_y, m_z are the moments acting about the x, y, z axes. The displacements and rotations for the end effector are represented thus:

$$\left[{}^{H}D\right] = \left[dx \;\; dy \;\; dz \;\; \delta x \;\; \delta y \;\; \delta z\right]^T \tag{9.35}$$

Similar elements can be defined for each joint i.e.

$$\left[T\right] = \left[T_1 \;\; T_2 \;\; T_3 \;\; T_4 \;\; T_5 \;\; T_6\right]^T \tag{9.36}$$

Here the T components represent the forces and torques for prismatic and revolute joints respectively. The displacements and rotations for the joints are represented thus:

$$\left[D_\theta\right] = \left[d\theta_1 \quad d\theta_2 \quad d\theta_3 \quad d\theta_4 \quad d\theta_5 \quad d\theta_6\right]^T \tag{9.37}$$

The amount of work done at the end effector must be the same as the combined work done at the joints [1] i.e.

$$\delta W = \left[{}^{H}F\right]^T \left[{}^{H}D\right] = \left[T\right]^T \left[D_\theta\right] \tag{9.38}$$

When the values are substituted the following equation is derived:

$$f_x dx + f_y dy + f_z dz + m_x \delta x + m_y \delta y + m_z \delta z = T_1 d\theta_1 + T_2 d\theta_2 + T_3 d\theta_3 + T_4 d\theta_4 + T_5 d\theta_5 + T_6 d\theta_6 \tag{9.39}$$

Previously, the Jacobian was used to relate differential motions. Using this relationship gives:

$$
\begin{aligned}
\left[{}^{H}F\right]^T \left[{}^{H}J\right]\left[D_\theta\right] &= \left[T\right]^T \left[D_\theta\right] \\
\therefore \left[{}^{H}F\right]^T \left[{}^{H}J\right] &= \left[T\right]^T
\end{aligned}
\tag{9.40}
$$

For these matrices the force/torque values for the joints can be calculated as follows:

$$\left[T\right] = \left[{}^{H}J\right]^T \left[{}^{H}F\right] \tag{9.41}$$

Here the Jacobian is calculated for the differential motion analysis. This allows the joint forces to be calculated for specific end effector forces/torques [1].

9.7 Force and Torque Transformations between Coordinate Frames

If a force acts on a link then its effects are experienced relative to the reference frames that are positioned either end of the link (i.e. at each joint). Therefore the transformation of the forces from one frame to the other must be considered. We define the forces/torques for a reference frame as [1]

$$[F] = [f_x \quad f_y \quad f_z \quad m_x \quad m_y \quad m_z]^T \tag{9.42}$$

$$[D] = [dx \quad dy \quad dz \quad \delta x \quad \delta y \quad \delta z]^T \tag{9.43}$$

The effects of the forces of the frame B would be ${}^BF$ and the displacements caused by the force are ${}^BD$. These are represented in the following way:

$$[{}^BF] = \left[{}^Bf_x \quad {}^Bf_y \quad {}^Bf_z \quad {}^Bm_x \quad {}^Bm_y \quad {}^Bm_z\right]^T \tag{9.44}$$

$$[{}^BD] = [{}^Bdx \quad {}^Bdy \quad {}^Bdz \quad {}^B\delta x \quad {}^B\delta y \quad {}^B\delta z]^T \tag{9.45}$$

The total virtual work done will be the same since both reference frames are connected to the link i.e

$$\delta W = [F]^T[D] = [{}^BF]^T[{}^BD] \tag{9.46}$$

It has been shown [3] that

$$[{}^BD] = [{}^BJ][D] \tag{9.47}$$

Therefore, Equation (9.46) becomes:

$$\begin{aligned} &[F]^T[D] = [{}^BF]^T[{}^BJ][D] \\ &\therefore [F]^T = [{}^BF]^T[{}^BJ] \end{aligned} \tag{9.48}$$

Hence

$$[{}^BF] = [{}^BJ]^T[F] \tag{9.49}$$

It has been found that it is easier to calculate the elements of F^B using the following relationships rather than calculating the Jacobian [3].

$$\begin{aligned} {}^Bf_x &= \mathbf{n} \bullet \mathbf{f} \\ {}^Bf_y &= \mathbf{o} \bullet \mathbf{f} \\ {}^Bf_z &= \mathbf{a} \bullet \mathbf{f} \\ {}^Bm_x &= \mathbf{n} \bullet [(\mathbf{f} \times \mathbf{p}) + \mathbf{m}] \\ {}^Bm_y &= \mathbf{o} \bullet [(\mathbf{f} \times \mathbf{p}) + \mathbf{m}] \\ {}^Bm_z &= \mathbf{a} \bullet [(\mathbf{f} \times \mathbf{p}) + \mathbf{m}] \end{aligned} \tag{9.50}$$

These equations enable the forces and torques to be defined for different reference frames.

References

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[3] Paul, R.P., *Robot Manipulators, Mathematics, Programming and Control*, The MIT Press, 1981.